



Towards Dynamic Interleaving Optimizer

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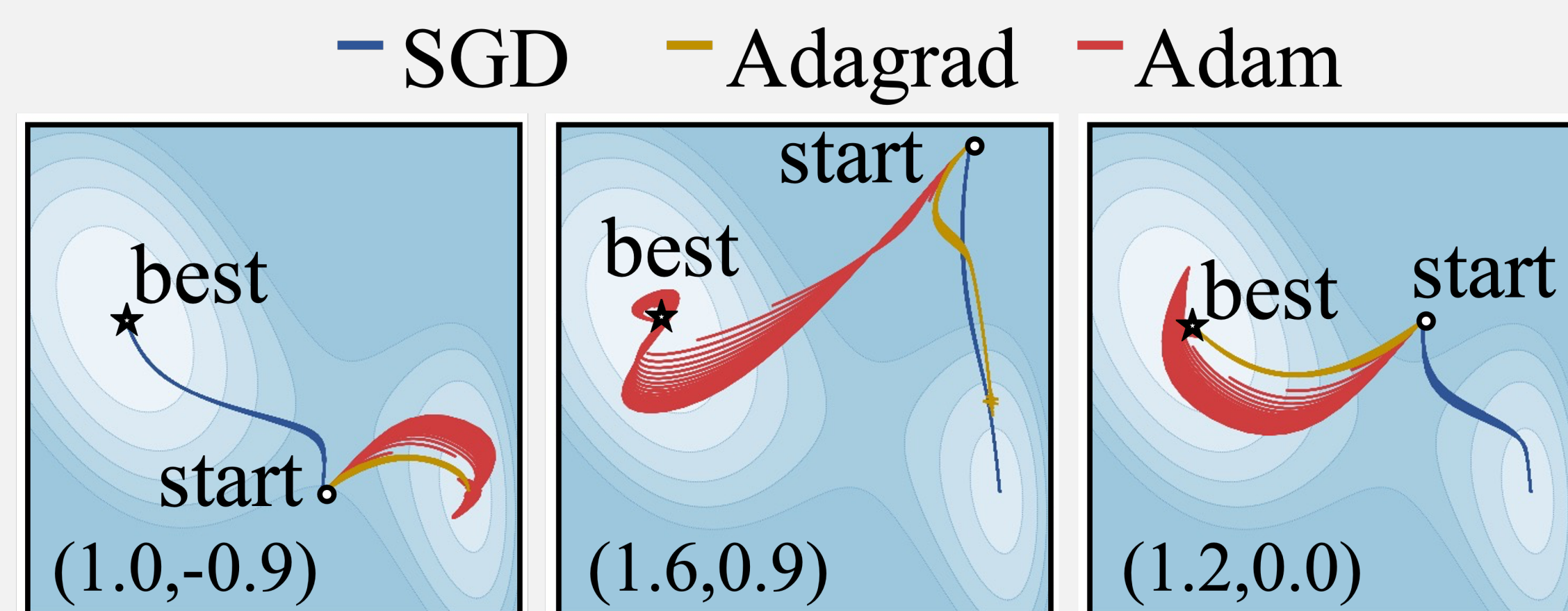


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Background and Motivation

State-Dependent Optimizer Suitability.

Different optimizers are suitable for different training stages or initial parameter states. For example, adaptive methods (e.g., Adam) performs well in early noisy regions, while SGD provides superior later refinement.



Optimal optimizers vary across different initial states.

Current Static or Hybrid Methods.

Current practices typically employ a global static optimizer or simple one-way switching (e.g., Adam-to-SGD), failing to adapt to the complex shifts in the loss landscape demands.

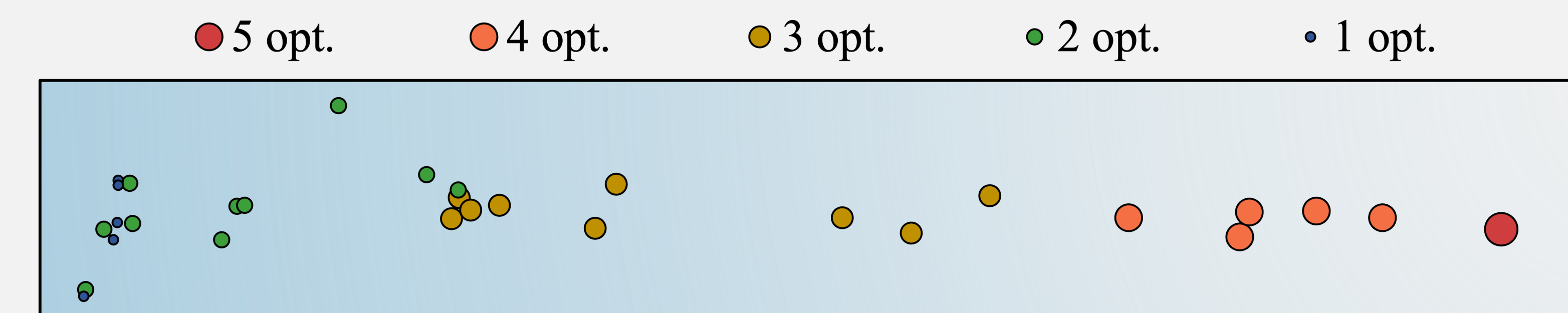
Thus, ...

- **Stagnated Convergence:** Static selection traps models in suboptimal regimes where the current optimizer's efficiency has already plateaued..
- **Optimization Bottlenecks:** Lack of bidirectional switching prevents models from escaping local sub-basins that require specific gradient behaviors.

Discussion

Overall Performance.

- Achieving superior accuracy by unlocking novel optimization paths through multi-strategy synergy.



Time Overhead.

- Our method introduces limited time overhead.

	forward	gradient	optimizer	transferability	surrogate	acquisition
FLOPs	2.2×10^{16}	4.4×10^{16}	$0.2 \sim 2.0 \times 10^{13}$	2.2×10^{13}	5.3×10^{14}	1.0×10^{14}
Proportion	33.3%	66.7%	quite small	0.3%	8.0%	1.5%

Methodology: Dynamic Optimizer Interleaving Training (DOIT)

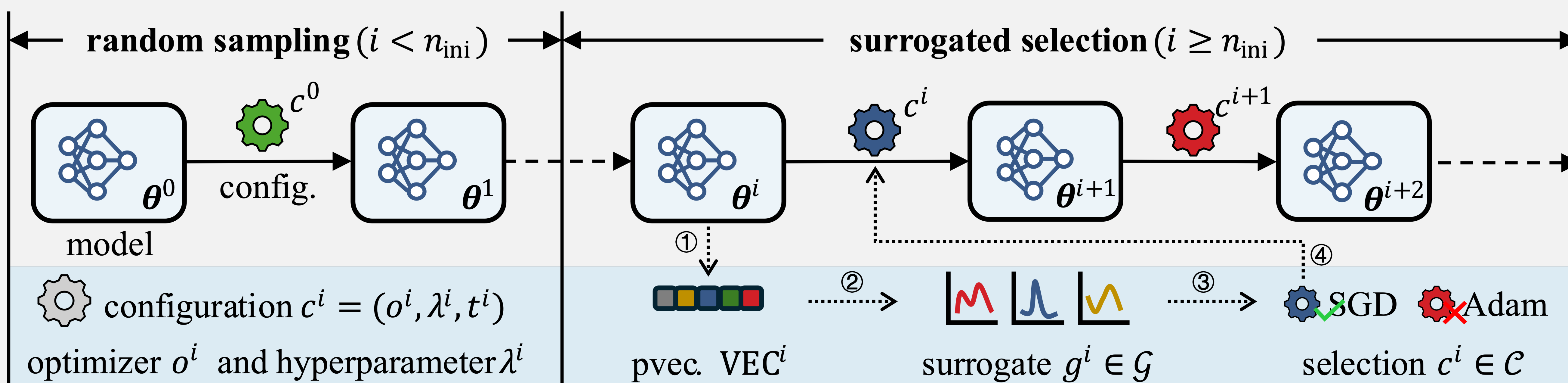
Surrogated Model.

random sampling + surrogated selection

- **Model:** Gaussian Process (GP).
- **Input:** model parameter VEC^i + Hyperparameter λ^i .
- **Output:** estimated performance score,
 $s = \tanh((\mu_\Delta + \Delta_{\text{UPPER}} + \Delta_{\text{LOWER}})/3 + \alpha\sigma_\Delta)$.

Acquisition Function.

- **Variance:** $ACQ(s_\mu, s_\sigma) = s_\mu + \alpha s_\sigma$.
- **Transferability:**
 $ACQ(s_\mu, s_\sigma, \omega_t) = s_\mu + \alpha(1 - \omega_t)s_\sigma$.
- **Training process:**
 $e = \text{sigmoid}(s_\mu + \alpha(1 - 2^{-\lfloor i/n \rfloor} \cdot \omega_t)s_\sigma)$.



Training process of our proposed Dynamic Optimizer Interleaving Training.

Experimental Results

Overall Performance.

- **Accuracy:** 1%-3% improvement with lower variance.

	usps	mnist	full training stl10	cifar10	imagenet	usps	stl10	mrpc	qqp
SGD	97.46 $\pm$ 0.70	99.50 $\pm$ 0.04	97.83 $\pm$ 0.22	97.53 $\pm$ 0.03	78.73 $\pm$ 0.38	94.42 $\pm$ 0.21	97.75 $\pm$ 0.16	85.21 $\pm$ 0.35	82.13 $\pm$ 0.52
SGDM	97.68 $\pm$ 0.11	99.65 $\pm$ 0.04	96.61 $\pm$ 0.04	97.58 $\pm$ 0.04	78.36 $\pm$ 0.12	95.67 $\pm$ 0.14	98.37 $\pm$ 0.10	85.54 $\pm$ 0.69	83.30 $\pm$ 0.63
Adagrad	93.40 $\pm$ 0.04	98.24 $\pm$ 0.50	78.66 $\pm$ 2.54	60.95 $\pm$ 0.62	77.59 $\pm$ 0.03	95.37 $\pm$ 0.21	98.34 $\pm$ 0.09	84.94 $\pm$ 0.59	83.47 $\pm$ 0.79
RMSprop	95.25 $\pm$ 0.04	98.14 $\pm$ 0.09	88.62 $\pm$ 4.45	78.09 $\pm$ 0.92	74.01 $\pm$ 0.08	94.64 $\pm$ 0.53	97.91 $\pm$ 0.07	84.09 $\pm$ 0.76	82.09 $\pm$ 0.63
Adam	93.26 $\pm$ 0.78	99.01 $\pm$ 0.08	82.26 $\pm$ 1.16	75.23 $\pm$ 0.92	73.96 $\pm$ 0.11	94.47 $\pm$ 0.49	98.36 $\pm$ 0.03	86.52 $\pm$ 0.71	82.27 $\pm$ 0.71
SWATS	94.00 $\pm$ 1.23	98.73 $\pm$ 0.13	88.03 $\pm$ 0.44	66.15 $\pm$ 4.74	76.93 $\pm$ 0.19	95.12 $\pm$ 0.21	98.38 $\pm$ 0.10	86.27 $\pm$ 0.62	80.79 $\pm$ 0.81
Padam	97.58 $\pm$ 0.11	99.66 $\pm$ 0.04	90.81 $\pm$ 0.06	96.03 $\pm$ 0.03	77.47 $\pm$ 0.16	95.72 $\pm$ 0.42	98.38 $\pm$ 0.10	80.64 $\pm$ 0.32	73.04 $\pm$ 0.91
AdaBound	87.64 $\pm$ 0.84	97.54 $\pm$ 0.13	86.33 $\pm$ 2.39	70.91 $\pm$ 4.35	75.93 $\pm$ 0.13	95.42 $\pm$ 0.11	98.30 $\pm$ 0.14	68.38 $\pm$ 0.59	78.64 $\pm$ 0.49
ours	97.81$\pm$0.21	99.71$\pm$0.02	98.21$\pm$0.19	98.04$\pm$0.03	79.98$\pm$0.01	96.12$\pm$0.10	99.01$\pm$0.09	87.99$\pm$0.13	85.57$\pm$0.14

- **Convergence:** faster with minimal computational cost.



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